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Review

Linking Microbiota to Human Diseases: A Systems Biology Perspective

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The human gut microbiota encompasses a densely populated ecosystem that provides essential functions for host development, immune maturation, and metabolism. Alterations to the gut microbiota have been observed in numerous diseases, including human metabolic diseases such as obesity, type 2 diabetes (T2D), and irritable bowel syndrome, and some animal experiments have suggested causality. However, few studies have validated causality in humans and the underlying mechanisms remain largely to be elucidated. We discuss how systems biology approaches combined with new experimental technologies may disentangle some of the mechanistic details in the complex interactions of diet, microbiota, and host metabolism and may provide testable hypotheses for advancing our current understanding of human–microbiota interaction.

Gut Microbiota in Human Health and Disease

Trillions of bacteria inhabit the human gastrointestinal tract and numerous studies have suggested that this ecosystem plays a major role in host physiology by affecting several processes ranging from maturation of the immune system, regulation of host metabolism, nutrition, and transformation of bioactive molecules and drugs to behavior [1]. Due to the emergence of systems biology approaches and new computational tools, there has been increasing interest in exploring the gut **microbiota** (see Glossary) and their combined genomes, the **microbiome**, as diagnostic and therapeutic targets for human diseases [2,3].

Compositional and functional alterations of the human gut microbiome have been linked to malnutrition [4] as well as obesity [5,6] and obesity-related diseases such as cirrhosis [7], cardiovascular disease (CVD) [8], and T2D [9,10]. Moreover, alterations of the gut microbiome have also been linked to inflammatory bowel disease (IBD) [11], colorectal cancer [12], neurodevelopmental disorders [13], and aging [14]. Considering the increasing global incidence of many of these conditions, changes in lifestyle and diet in the post-industrialization/westernization era have been argued to contribute to their emergence by shifting gut microbiota ecology [15].

Obesity, T2D, and IBD are characterized by reduced fecal **microbial diversity** [5,6,10,16] and recent studies have demonstrated that the consumption of artificial sweeteners and dietary emulsifier alters the gut microbiota, resulting in intestinal inflammation and development of the metabolic syndrome [17,18]. These findings suggest that the modern lifestyle has the potential to affect the gut microbiota, which may contribute to disease development. By contrast, societies with traditional lifestyles, such as hunter–gatherers, are signified by high microbial diversity [19,20]. Disruption of the microbial ecosystem by antibiotics early in life has been linked

Trends

The gut microbiota is altered in human metabolic diseases. Emerging data show that bacterial functions linked to colonic butyrate production are consistently associated with improved insulin sensitivity, indicating new possible therapeutic avenues.

Human metabolic diseases are often associated with decreased diversity and functional richness of the gut microbiota. Modern lifestyles characterized by the widespread use of antimicrobials and consumption of energy-dense foods, additives, and emulsifiers might contribute to loss of microbiota diversity and the increased incidence of chronic diseases.

Integration of patient stratification with microbiome functional profiling will be fundamental for the development of personalized medicine. Personal microbiome profiles will be important determinants of the effects of diet and the efficacy of therapeutic drugs.

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to the development of obesity in humans [21], and mouse studies demonstrated that early administration of antibiotics decreases ileal expression of host immune genes by depleting the microbiota and induces obesity in adulthood, potentially by affecting host liver metabolism [22,23]. As the microbiota matures during the first years of life, with the main changes occurring within the first 12 months [24–26], it is likely that microbiota perturbations during postnatal life may be critical for the host's long-term metabolic regulation by affecting the immune system, which co-develops with the gut microbiota. Therefore, understanding factors that shape the gut microbiota might lead to new therapies for both metabolic and inflammatory diseases.

In this review we discuss the gut microbiota with particular attention to metabolic diseases and we aim to summarize the most recent findings for how an altered gut microbiota may contribute to metabolic diseases and how systems biology approaches can be applied to study the complex interactions between the microbiota and its host.

Tripartite Interaction among Host, Diet, and Microbes: The Complexity and the Balance

Host–Microbe Interactions

Host–microbe interactions are determined by the complexity of the human body and the diversity of the microbiome (Figure 1). **Metagenomics** analysis of various body sites has revealed specific distributions of multi-kingdom communities (i.e., bacteria, viruses, and fungi) on the human body, with distinct genetic capacities reflecting the diversity of skin microenvironments [27]. Interactions between these kingdoms are also likely to occur in the human gut and recent studies have proposed a role for trans-kingdom interactions in the regulation of immune homeostasis and development of IBD [28]. Based on available techniques such as 16S rRNA gene sequencing, most studies have, however, so far focused on the characterization of bacteria in different habitats and on the identification of host responses to bacterial taxa and molecules.

The gut microbiota has a profound role in the development of intestinal lymphoid tissue [29] and can affect the balance between Th1 and Th2 responses [30] as well as induce Th17 responses, as proved for the commensal segmented filamentous bacteria [31,32]. The gut microbiota also affects the enteroendocrine system and has been shown to modulate gut hormones such as glucagon-like peptide 1 (GLP-1) [33,34], which decreases food intake and promotes insulin sensitivity [35] but also has important local effects in the gut including modulation of intraepithelial lymphocytes and regulation of gut transit [33,36]. A new study demonstrated that the gut microbiota modulates the development of the enteric nervous system (ENS) [37], which is essential for the regulation of gut transit. Neurons within the ENS express GLP-1 receptors but it is currently unclear whether the effects of GLP-1 on gut motility are mediated through the ENS. Furthermore, recent work in mice and humans revealed that the composition of the gut microbiota oscillates with the circadian clock [38], which also implies that it may be important to standardize sampling procedures.

Human genetics might play a role in shaping the composition of the gut microbiota but only a few studies have so far addressed this aspect of host–microbe interaction. One example is the observation of an altered microbiota in people homozygous for loss-of-function alleles of the *FUT2* gene [39]. *FUT2* encodes an enzyme required for the fucosylation of surface carbohydrates on mucosal linings and defects in FUT2 alter both the composition and the functionality of the gut microbiota, as demonstrated by combining microbiota profiling with metabolomics [39]. Interestingly, *FUT2* has been linked to Crohn's disease [40], suggesting that an altered gut microbiota may explain the association between *FUT2* genotype and increased susceptibility to Crohn's disease. The role of the human genotype in shaping the gut microbiota has also been investigated in the context of obesity and profiling of the fecal microbiota in a large twin cohort revealed that the abundance of several microbial taxa was influenced by host genetics and that

Glossary

Bile acids: primary bile acids [e.g., cholic acid (CA) in mice and humans, chenodeoxycholic acid (CDCA) in humans, muricholic acids in mice] are synthesized from cholesterol in the liver and conjugated with glycine or taurine. The gut microbiota deconjugates and transforms primary bile acids into secondary bile acids [e.g., deoxycholic acid (DCA), lithocholic acid (LCA)].

Fecal microbiota transplantation (FMT): a procedure in which stool samples from healthy donors are introduced to patients through either the lower or the upper gastrointestinal tract. FMT has proved successful for the treatment of *Clostridium difficile* infections.

Metagenomics: the application of genomic sequencing technologies to study the genetic material directly recovered from an environmental/clinical sample (e.g., the microbiome).

Microbial diversity: the number of microbial taxa that are represented within a given habitat/community. Many human chronic diseases (e.g., IBD, obesity) are characterized by decreased fecal microbial diversity.

Microbiome: the collection of the genes/genomes encoded by the microbiota.

Microbiota: the collection of microbial taxa in a given environment.

Prebiotics: dietary ingredients that are fermented by specific gut microbes and that result in specific changes in the composition and/or activity of the gastrointestinal microbiota, thus conferring benefits on host health[i].

Probiotics: live microorganisms that, when administered in adequate amounts, confer a health benefit on the host[i].

Short-chain fatty acids (SCFAs): end products of the fermentation of dietary carbohydrates and fiber by the colonic microbiota. Acetate, propionate, and butyrate are the most abundant SCFAs.



Figure 1. Complexity and Balance in the Interaction of the Gut Microbiota with the Host. Factors contributing to host–microbe interactions are indicated with different colors. Host factors include ethnicity or geography, critical developmental periods (infant), circadian rhythm, gene polymorphisms, epigenetics, body sites, enteric nervous system (ENS), diet, innate and adaptive immunity, bile acids (BAs), and host metabolites that can affect microbial colonization of the host [e.g., mucins, reactive oxygen species (ROS), nitrate]. Microbial factors include microbe-associated molecular patterns (MAMPs) [such as flagellin, lipopolysaccharide (LPS), and peptidoglycan (PGN)], short-chain fatty acids (SCFAs), viruses, fungi, and bacterial components of the microbiota. For bacteria, the five most abundant phyla – Firmicutes (Fir), Bacteroidetes (Bac), Actinobacteria (Act), Proteobacteria (Pro), and Verrucomicrobia (Ver) – are highlighted.

some of these taxa co-occurred in lean people [41], indicating an interaction between host genotype, gut microbiota, and metabolism. However, further investigations are needed to determine the extent to which host genotype impacts the microbiome, as studies in mice show that environmental factors might have bigger effects than genetics [42].

Diet–Microbiota Interactions

Energy intake and the macronutrient composition of the diet affect human health and impact the composition of the human gut microbiota [43]; in particular, recent mouse studies performed on strains of different genetic background show that diet has a stronger effect than host genotype in determining gut microbial composition [44]. The gut microbiota responds rapidly to dietary interventions [45–47] and short-term consumption of diets comprising entirely either animal or plant products can alter the overall structure of the gut microbiota [45]. However, individual

dietary preferences seem to affect the structure of the microbiota in the long term [47]. For example, the gut microbiota of people consuming high amounts of protein and animal fat is dominated by the *Bacteroides* genus, while the gut microbiota of people consuming more fiber and carbohydrates is dominated by the *Prevotella* genus [47]. Similar findings were observed in a study comparing children in Burkina Faso and Italy, demonstrating reduced levels of *Prevotella* and increased levels of *Bacteroides* in the Italians, which correlated with their lower fiber intake [48]. However, comparison of human omnivores and vegans did not show clear taxonomic shifts in the gut community, but revealed distinct profiles of bacterial metabolites in plasma [49], and reduced capacity to metabolize L-carnitine, present in red meats, to trimethylamine (TMA) [50]. These results indicate that microbial metabolism in omnivores and vegans differs despite similar microbial composition and suggest that the microbiota adapts to shifts in diet and may facilitate digestion of specific nutrients.

The notion that different dietary components can modulate the microbiota can also be used therapeutically. One promising strategy is the development and application of **prebiotics**, as strongly suggested by the health benefits of dietary fiber and of the metabolic end products of microbial fiber fermentation, such as **short-chain fatty acids (SCFAs)** [51,52]. Consumption of prebiotics and fiber can induce shifts in specific microbial groups and promote beneficial metabolic effects [53,54], but it should be noted that the composition of an individual's microbiota might determine the responsiveness to the dietary intervention [53,55].

Microbe–Microbe Interactions

Recent advances indicate that microbe–microbe interactions such as metabolic interactions, quorum sensing, and competition might also affect gut microbiota composition. For example, *Salmonella enterica* serovar Typhimurium and *Clostridium difficile*, two antibiotic-associated pathogens, improve their fitness after antibiotic treatment by utilizing microbiota-liberated mucosal carbohydrates such as sialic acid [56]. Additionally, the growth of *Bacteroides vulgatus* on inulin is made possible by the utilization of inulin metabolites produced by the primary degrader *Bacteroides ovatus* [57]. Furthermore, members of the Enterobacteriaceae caused weight loss in gnotobiotic mice fed a Malawian diet when combined with strains of Bacteroidetes isolated from undernourished Malawian children [58]. Recent studies also show that microbial group behaviors such as quorum sensing promote the expansion of microbial taxa in the Firmicutes after disruption of the gut microbiota by antibiotic treatment [59] and that microbial competition through production of bacteriocins might be important for fighting intestinal infections [60]. Together, these studies illustrate the importance of bacteria–bacteria and bacteria–diet interactions for human health. However, our knowledge about microbe–microbe and trans-kingdom interactions in the gut remains limited.

Gut Microbiota in Metabolic Diseases

Obesity

The obesity epidemic has become a major public health issue due to its links to related metabolic disorders such as T2D, CVD, and nonalcoholic fatty liver disease. Human genetics, diet, and sedentary lifestyle are risk factors for obesity while common treatments include calorie restriction, increased physical activity, antiobesity medications, and weight-loss (bariatric) surgery. Accumulating data have recently suggested that the gut microbiota may be another factor in the global obesity epidemic [61]. Initial evidence for the role of the gut microbiota in promoting adiposity comes from studies in germ-free mice showing resistance to diet-induced obesity [62] depending on the macronutrient composition of the diet [63] and increased fat storage on colonization with microbiota from conventional donors [64]. Subsequently, it was shown that transplantation of microbiota from both obese mice and humans promoted higher fat accumulation in recipient germ-free mice compared with transplantation of microbiota from lean donors [65,66]. Interestingly, a recent case report indicated significant weight gain in a woman receiving

fecal microbiota transplantation (FMT) from a healthy overweight donor as a treatment for *C. difficile* infection [67], suggesting that obesity may also be transmissible in humans. Overall, these studies have shown that the gut microbiota is an environmental factor promoting fat storage.

Recent studies have shown that obese people have reduced fecal microbiota richness [5,6], which is linked to increased inflammation and decreased insulin sensitivity [5]. Additionally, a large-scale twin study showed a connection between human genetics and the heritability of microbial taxa [41]. Specifically, the abundance of Christensenellaceae was highly correlated in monozygotic twin pairs and associated with a lean phenotype. FMT of an obese-associated microbiota supplemented with a culturable member of the Christensenellaceae resulted in reduced weight gain and altered gut microbiota composition in the recipient mice [41], demonstrating a clear relationship between heritable microbial taxa and host metabolic regulation. These studies highlight the possibility of developing bacterial taxa as novel **probiotics** targeting metabolic diseases. Future studies should focus on the identification of the genes that influence microbiota heritability and of associated host–microbe interactions promoting leanness.

Bariatric surgery is the most successful long-term treatment for obesity [68]. Several studies have demonstrated that one of the most effective bariatric surgery procedures, Roux-en-Y gastric bypass (RYGB), alters the human gut microbiota promoting an increase in Enterobacteriaceae and microbiota richness in the short term up to 6 months after surgery [69–72]. We recently found that changes to the gut microbiota were stable up to 9 years after surgery and that transplantation of the surgically altered microbiota resulted in reduced fat-mass gain in recipient germ-free mice [73]. Another bariatric surgery procedure, vertical sleeve gastrectomy (VSG), has recently been proved to cause less postoperative complications than RYGB but still achieve similar weight loss and amelioration of obesity-related comorbidities up to 3 years after surgery [74]. Importantly, both RYGB and VSG result in increased circulation of **bile acids** in humans and mice [75,76], and we recently found that the beneficial effects of VSG on fat-mass loss and glucose metabolism required intact signaling through the bile acid receptor FXR. We also observed that VSG induced changes in the gut microbiota, some of which were dependent on FXR [77]. However, it is at present unknown whether polymorphisms in *FXR* affect the outcome of bariatric surgery in humans and how an altered microbiota following bariatric surgery contributes to altered bile acid metabolism. Increased knowledge may lead to the development of less invasive approaches for the treatment of obesity and its comorbidities.

T2D

The gut microbiota is altered in T2D and specific microbiota functions such as production of butyrate and secondary bile acids may be important for insulin sensitivity. While genome-wide association studies revealed weak links between human genetic polymorphisms and T2D [78], the gut microbiome has successfully been used to classify T2D patients from healthy people in two independent studies of European and Chinese populations [9,10]. The algorithms used to classify T2D revealed that the microbial discriminant markers were different in the two populations; however, T2D was associated with high levels of *Lactobacillus* and low levels of *Roseburia* in both T2D cohorts [9], showing that some consistent features can be linked to T2D independent of geographical and dietary differences. In addition, FMT of lean microbiota into patients with metabolic syndrome improved insulin sensitivity over a 6-week period and was associated with increased levels of *Roseburia* in the fecal microbiota and of another butyrate producer, *Eubacterium hallii*, in the microbiota from the small intestinal mucosa [79]. Besides butyrate producers, other Gram-positive bacteria have been linked to insulin sensitivity: treatment of metabolic syndrome patients with vancomycin, an antibiotic directed against Gram-positive bacteria, resulted in decreased insulin sensitivity and reduced levels of plasma secondary bile acids [80]. Therefore, these initial human studies indicate that supplementation with Gram-positive bacteria associated with butyrate production and bile acid metabolism may

provide new treatment opportunities for metabolic diseases, but additional work including prospective longitudinal studies is required for validation of the initial findings.

Mechanistic understanding of microbial effects on host metabolism is so far limited to germ-free mice, which have demonstrated that the gut microbiota affects almost all organ systems that contribute to metabolic control: the gut microbiota modulates appetite and food intake [64,81], absorption of nutrients from the gut [64,82], hepatic steatosis [83], inflammation and triglyceride accumulation in adipose tissue [64,84,85], and fatty acid oxidation in skeletal muscle and liver [62]. However, there is limited knowledge on how the gut microbiota affects human metabolism. One potential approach would be to collect biopsies and use systems biology approaches to identify microbial taxa, their genes, and transcriptional responses that interact with host signaling pathways. Although these studies do not provide direct causal evidence, they will improve our understanding of human–microbe interactions.

CVD

CVD is one of the leading causes of deaths worldwide. Initial evidence for a role of microbes in CVD came from association studies between atherosclerosis and infectious agents [86] and between CVD and impaired dental health [87]. Moreover, statins, which are the first-line treatment for the prevention of CVD, have been shown to have antimicrobial effects [88], and plasma levels of several secondary bile acids were associated with a good response to simvastatin treatment [89]. Therefore, although the effects of statins on gut microbiota composition and function are unknown, the gut microbiota may contribute to CVD and play a role in determining the therapeutic response to statins. The effect of microbial metabolism on therapeutic drugs is illustrated in Box 1.

Recently, we and others have identified bacterial DNA in atherosclerotic plaques and observed that bacterial load within plaques may contribute to their stability and CVD events [90,91]. Metagenomic analysis of the fecal microbiome in patients with symptomatic atherosclerosis and healthy controls revealed compositional and functional alterations comprising decreased levels of genes for the production of β-carotene, which is anti-inflammatory, and increased levels of genes for the synthesis of peptidoglycan, which is proinflammatory, in the gut microbiome of atherosclerosis patients [92]. Additional work has shown that microbial metabolism of dietary phosphatidylcholine and L-carnitine to TMA may be involved in atherosclerosis and CVD, through a mechanism involving oxidation of TMA to TMA-*N*-oxide (TMAO) in the liver [50,93]. TMAO was strongly associated with CVD and death [94] and a causal role for TMA/TMAO metabolism in atherosclerosis was demonstrated by the suppression of TMAO production in atherosclerosis-prone *Apoe*$^{-/-}$ mice fed L-carnitine and choline diets and treated with

Box 1. Microbial Metabolism of Drugs

The gut microbiome encodes three orders of magnitude more genes than the human genome [119] and the microbial capacity to transform foreign compounds (e.g., xenobiotics) may affect host responses to therapeutic drugs and influence their efficacy [120]. Recent evidence supports that the effects of metformin, the most prescribed antidiabetic drug, might be regulated by host-associated microbes: the increase in lifespan induced by metformin in *Caenorhabditis elegans* was mediated by disruption of folate and methionine cycles in the co-cultured *Escherichia coli*, resulting in methionine nutritional restriction for the worm [121]. In addition, the improvement of glucose metabolism by metformin treatment in high-fat-fed mice was associated with increased levels of *Akkermansia* [122]. Also, the gut microbiota of European women with T2D treated with metformin was enriched in Enterobacteriaceae [9], but it is currently unknown whether metformin acts through altering the human gut microbiota and its function.

Another example of the effects of microbial metabolism on therapeutic drugs is the inactivation of the cardiac drug digoxin by the gut microbe *Eggerthella lenta*, which can reduce digoxin to its inactive metabolite dihydrodigoxin [123]. Together these studies highlight the possibility that gut microbiota composition and metabolism may modulate the therapeutic effects of drugs and indicate that understanding the interactions between microbiota, diet, and host might also be of value in the context of pharmacology.

antibiotics [50,93]. Therefore, microbial metabolism in combination with diet interactions may be involved in the development of CVD.

New Experimental and Computational Approaches for Understanding Host–Microbe–Diet Interactions

Systems biology approaches integrating 'omics' techniques (Figure 2, Key Figure) can be used to unravel the mechanisms behind host–microbe interactions. Omics techniques allow the study of the collective genomes and taxonomic groups (metagenomics), transcriptomes

Key Figure

Integration of 'Omics' Techniques

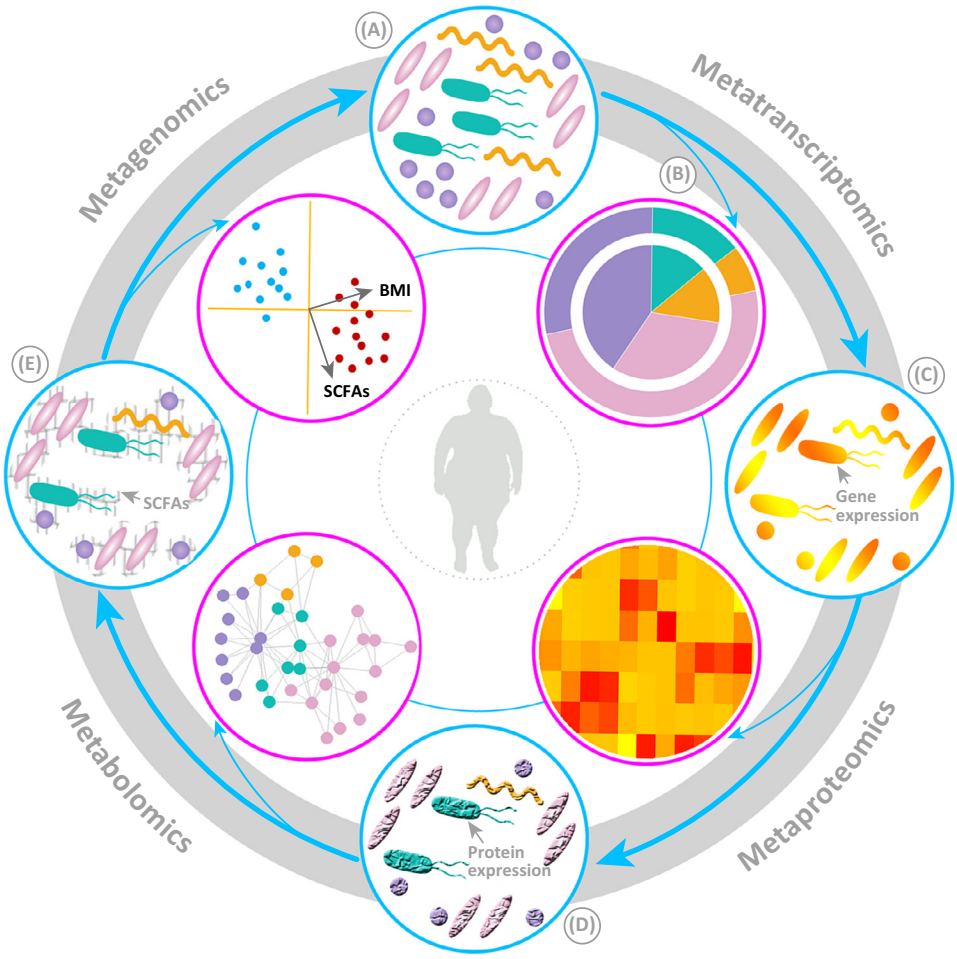


Figure 2. Due to the complexity of host–microbe–diet interactions, a number of omics systems biology approaches need to be integrated to reveal the mechanistic roles of the gut microbiota and provide diagnostic and therapeutic methods for treating human diseases. Gut microbiota alterations (A) can be detected as taxonomic changes or differences in the microbiome (B) by means of metagenomics techniques. Bioinformatic analyses can be applied to assess variations in (C) microbial gene expression (metatranscriptomics), (D) protein expression (metaproteomics), and (E) metabolism [e.g., short-chain fatty acids (SCFAs); metabolomics] linked to gut microbiota alterations.

(metatranscriptomics), proteomes (metaproteomics), and metabolomes (metabolomics) of the gut microbiota. Taxonomic and metagenomic profiling based on whole-genome shotgun sequencing can help in understanding 'which microbes are in the community' and 'what they are capable of doing'. However, the overlap between metagenomics and metatranscriptomics was only 41% [95], illustrating the importance of moving beyond metagenomics to understand 'what the community is really doing'. Furthermore, given the importance of post-transcriptional and translational regulation and the fact that protein abundance is not fully coupled to regulation of gene expression in either eukaryotes or bacteria [96,97], metaproteomics and metabolomics should be applied to accurately characterize the functional capacities of a microbial community. However, lack of sequenced microbial genomes is still limiting the annotation and interpretation of metatranscriptomic and proteomic data.

By combining different omics techniques and using systems biology approaches, researchers are increasing our understanding of microbial contributions to human health. For example, the comparison of human gut metagenomes and metatranscriptomes revealed not only that gene expression is consistent with DNA abundance for less than 50% of the genes in the microbiome, but also that some genes (e.g., genes involved in the citrate cycle) are overexpressed compared with their DNA levels while others (e.g., genes in starch and sucrose metabolism) are under-expressed [95]. The authors also observe that, depending on the subject, the relative abundance in the metatranscriptome of transcripts from the archaeon *Methanobrevibacter smithii* (2.1–14.7%) was higher than its DNA abundance in the metagenome (0.5–5.3%) [95]. These results may help us to better understand the energy harvesting capability of this archaeon and its link with obesity [98]. In this context it will be important to investigate how individual metagenome–metatranscriptome correlations change on diet interventions.

Another study comparing the metagenome and metatranscriptome of oral microbiota in healthy subjects and patients with periodontitis confirmed the inconsistency between gene abundance and expression. This study demonstrated that many bacteria, such as *Bacteroidetes* oral taxa 274 and *Aggregatibacter actinomycetemcomitans*, can upregulate or downregulate their transcriptomes in response to disease state without major shifts in genome abundance [99], while some bacteria highly abundant in periodontitis, such as *Lactococcus lactis*, might be metabolically inactive [99]. Interestingly, most of the virulence factors overexpressed in periodontitis belonged to bacteria that had not been previously linked to pathogenesis [99]. These studies highlight the usefulness of combining metatranscriptomics and metagenomics to identify microbial functions potentially relevant for the pathophysiology of a disease.

Recently, new approaches have been proposed for the analysis of shotgun metagenome data that might provide a more dynamic description of the ecosystem and lead to the identification of bacteria active in human diseases. One approach is based on the assumption that bacterial cells in a growing population will be at different stages of DNA replication and produce a sequencing coverage pattern with a single trough and a single peak near the replication origin, while bacteria in a non-dividing population will produce a flat sequencing coverage pattern along the genome [100]. This study suggested that growth rates estimated by peak-to-trough coverage ratios recovered from metagenomics data are better correlated with IBD and T2D than the relative abundance of DNA [100]. Therefore, the development of new bioinformatics approaches is fundamental for improving the utilization of the vast amount of metagenomics data that has been produced in recent years.

Despite its usefulness in describing microbial functions that are actively expressed, metatranscriptomics provides only a snapshot of the dynamic interactions between host, gut microbes, and environment, since most bacterial mRNAs have a short half-life of minutes [101]. Therefore, metaproteomics has been applied in a few proof-of-principle studies to retrieve disease markers.

The combination of metagenomics and metaproteomics identified metagenomics markers and protein signals for Crohn's disease [102,103]. Furthermore, metaproteomics and taxonomic profiles of obese and non-obese individuals demonstrated that, despite a general low abundance, Bacteroidetes were metabolically more active in obese people [104], offering an alternative perspective on the association between Bacteroidetes and obesity. However, large-scale metaproteomics analysis is not free of limitations as it requires laborious sample preparation and extensive data processing, while its results are dependent on well-annotated reference gene/protein databases and the availability of state-of-the-art mass spectrometry instruments [105,106].

A complimentary approach for the study of microbiota functionality and host–microbiota co-metabolism is provided by metabolomics, which analyzes the small-molecule composition of host fluids and tissues to detect metabolites derived from bacteria or organisms other than the host [107]. This approach has shown that several microbial metabolites are associated with metabolic syndrome in humans and experimental models [108]. Targeted metabolomics can also be applied to test whether shifts in microbial gene functions are coupled to shifts in community functionality, as has been done by profiling SCFAs and bile acids in obesity and after bariatric surgery-mediated weight loss [73,77]. In summary, integration of metabolomics with metagenomics, metatranscriptomics, or metaproteomics might be required for a deeper understanding of host–microbe interactions and co-metabolism as well as to increase our understanding of diet-microbiota interactions.

Although extremely useful, omics techniques often produce correlative evidence for a link between shifts in the microbiome and human diseases. To provide causal evidence, new

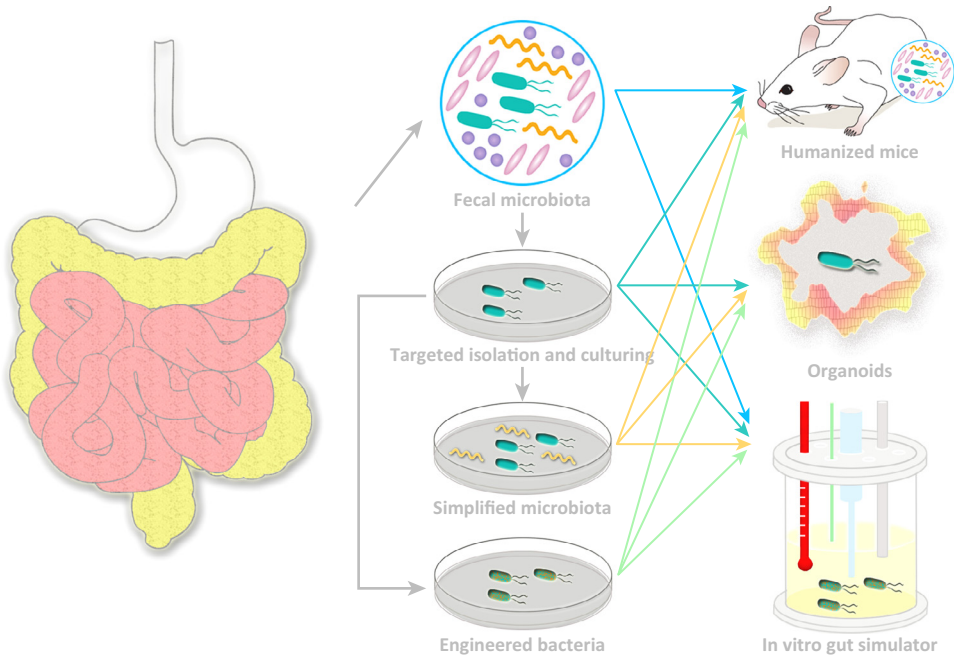


Figure 3. Emerging New Technologies to Investigate the Causal Relationship between Microbiota and Human Diseases. Transplantation of human fecal microbiota, isolation of human intestinal bacteria, simplified microbiota, and engineered bacteria in combination with the germ-free mouse model, on different genetic backgrounds or fed diets with different macronutrient compositions, can be used to investigate human–microbe–diet interactions. *In vitro* gut simulators and co-culturing of bacteria and/or bacterial metabolites with organoids might also add mechanistic knowledge on microbe–microbe and host–microbe interactions.

experimental methods are being developed that will help in investigating whether an altered gut microbiota may translate into altered metabolism and physiology (Figure 3). Humanized mice have proved to be of great value for studying host metabolic and immune phenotypes [109,110]. The development of simplified communities comprising human intestinal microbes [111] and/or engineered bacteria could also bring new knowledge on the mechanisms underlying host–microbe–diet interactions. For example, the administration of *Escherichia coli* Nissle 1917 synthesizing precursors of anorexigenic lipids reduced food intake and adiposity in mice fed a high-fat diet [112], while lactobacilli engineered to secrete full-length GLP-1 increased glucose tolerance in diabetic rats [113]. Compositional and metabolic shifts of the gut microbiota caused by changes in diet and intestinal physicochemical parameters might also be studied in *in vitro* gut simulators [114], while co-culturing of organoids with bacterial metabolites or isolated bacteria might help our understanding of the roles of host–microbe and host–pathogen interactions in specific tissues. Using ileal organoids, one recent study demonstrated a strong effect of the human intestinal bacterium *Akkermansia muciniphia* on the expression of host genes for lipid metabolism [115].

Concluding Remarks and Future Perspectives

Animal studies have indicated a direct role for the gut microbiota in the development of several diseases, such as obesity and colitis [66,116]. However, solid and consistent evidence for the contribution of the gut microbiota to human diseases is lacking due to the vast interindividual variability of the human gut microbiome [117], the potential confounding effects of diet and medications, and differences in the pipelines for gut microbiota analyses. Large prospective studies may help in determining causal relationships between the gut microbiota and various diseases (see Outstanding Questions).

As a new, emerging field, multiple challenges need to be addressed to expand our current understanding of the human gut microbiota. For example, although mice provide a valuable model, transfer of human microbiota does not result in full induction of the murine immune system [32] and humans and mice have different bile acid compositions, which may complicate the translation of results. It is also unclear to what extent the fecal microbiota reflects the microbiota of different niches within the gut: a recent study based on mucosal biopsies revealed distinct microbial populations in the distal colon and in fecal samples [118], suggesting that the fecal microbiota might predominantly reflect the composition of the luminal microbiota and that the functions of microbes specifically residing in intestinal subniches might be largely unknown.

To identify microbial taxa and gene functions consistently associated with human diseases it will be essential to standardize the analytical procedures and take into consideration the multiplicity of genetic, geographical, and dietary factors shaping the human gut microbiota. We also need to improve our catalogs of cultured human gut microbes and allow their functional characterization. These approaches may provide novel probiotics strains for the treatment of metabolic diseases and possibilities for personalized medicine that integrates the personal microbiome, supplementation with specific bacteria, and dietary patterns.

Outstanding Questions

What is a healthy microbiome? What are the roles of trans-kingdom interactions between bacteria, viruses, and fungi for immune and metabolic homeostasis? Longitudinal studies are required to characterize the dynamic changes of the normal gut microbiota.

How can we correct an aberrant gut microbiota to promote health and prevent disease?

What are the mechanistic links between an altered gut microbiota and a specific human disease? Systems biology (omics) approaches need to be integrated to expand current knowledge.

What is the composition of the microbiome in defined microniches within the human intestine? Do site-specific communities play a role in human–microbe interactions in health and disease?

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Resources

[i] www.isapp.net/

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